Highlights

A Data-Driven High-Fidelity Simulation Framework for Validating Online Adaptive PID Control in Differential-Drive Mobile Robots

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- A physics-calibrated simulation framework using CoppeliaSim-derived parameters
- Error-driven adaptive PID algorithm with O(500) computational complexity per cycle
- Comprehensive friction model including Coulomb, Stribeck, and viscous components
- Validation across five case studies with severe parameter variations (10× friction, 120% mass)
- Over 90% tracking error reduction compared to optimally-tuned fixed-gain PID

A Data-Driven High-Fidelity Simulation Framework for Validating Online Adaptive PID Control in Differential-Drive Mobile Robots

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Abstract

This paper presents a data-driven high-fidelity simulation framework for developing and validating adaptive control algorithms for differential-drive wheeled mobile robots. The simulation environment integrates a Python-based forward dynamics engine calibrated using physical parameters extracted from CoppeliaSim's Bullet physics engine, ensuring high physical fidelity while maintaining computational efficiency. Within this framework, we propose an error-driven adaptive PID gain scheduling algorithm that dynamically adjusts proportional, integral, and derivative gains based on real-time tracking error statistics—without requiring system model knowledge or offline optimization. The simulation incorporates realistic actuator dynamics (DC motor electrical/mechanical coupling), battery voltage sag under load, and a comprehensive Coulomb-Viscous-Stribeck friction model. Validation across five case studies demonstrates that the adaptive algorithm achieves over 90% reduction in position tracking error under severe model mismatch

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(10× friction increase) and 79% improvement in complex multi-disturbance scenarios (120% payload increase, surface transitions). The framework's modular architecture enables rapid prototyping of control algorithms with confidence in sim-to-real transferability. We provide theoretical stability analysis and discuss measures taken to address the simulation-to-reality gap, including domain randomization and actuator saturation modeling.

Keywords: simulation framework, model validation, adaptive control, mobile robot simulation, physics-based modeling, PID tuning, sim-to-real transfer

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1. Introduction

Trajectory tracking control is fundamental to autonomous mobile robotics, yet a persistent gap remains between theoretical advances and practical deployment [1, 2]. Differential-drive wheeled mobile robots (DDWMRs) are widely used in logistics, service robotics, and research due to their mechanical simplicity and maneuverability. However, achieving robust trajectory tracking under real-world conditions—where friction coefficients vary, payloads change dynamically, and actuator characteristics drift—remains an open challenge.

The standard industrial approach employs fixed-gain Proportional-Integral-Derivative (PID) controllers, valued for their simplicity and decades of proven reliability. Yet fixed-gain controllers suffer from a fundamental limitation: gains optimized for one operating condition inevitably degrade when conditions change. A controller tuned for an empty robot on smooth flooring may exhibit oscillations when payload increases or tracking lag when traversing rough terrain. This forces practitioners to either (i) tune conservatively, sacrificing performance for robustness, or (ii) re-tune for each deployment scenario, incurring significant engineering cost.

Advanced control methods address this limitation through adaptive or model-predictive approaches. Model Predictive Control (MPC) [3] can handle constraints and preview future trajectories, but requires solving optimization problems at each control cycle—computationally prohibitive for embedded platforms operating at 1–10 kHz. Model Reference Adaptive Control (MRAC) [4, 5] adjusts controller parameters online but typically requires accurate system models and persistence of excitation conditions. Reinforcement learning (RL) approaches [6, 7] can learn adaptive policies but demand extensive offline training and face sim-to-real transfer challenges [8, 9].

Gain scheduling offers a middle ground, adapting controller gains based on measurable operating conditions [10, 11]. Traditional gain scheduling, however, relies on offline computation: designers linearize the plant at multiple operating points, solve Linear Matrix Inequality (LMI) problems to ensure stability [12, 13], and interpolate gains during runtime. This approach is inflexible for unexpected trajectories and cannot learn from online experience. Self-tuning regulators [14] and Iterative Learning Control (ILC) [15] provide some online capability, but the former imposes high computational cost while the latter requires trajectory repetition—unsuitable for non-repetitive missions.

Recent theoretical developments suggest a path forward. Model-Free Adaptive Control (MFAC) [16] demonstrates that data-driven methods can achieve excellent performance without explicit system identification. Lyapunov-based analysis proves that adaptive PID controllers can be rigorously stable even for uncertain nonlinear systems [17]. Active Disturbance Rejection Control (ADRC) [18, 19] treats all uncertainties as a "total disturbance" estimated and canceled in real-time. Dual-loop architectures [20, 21] decouple kinematic and dynamic control, simplifying the adaptation problem. Studies on payload variation [22] confirm that adaptive compensation is essential for maintaining tracking accuracy under changing inertial properties.

Despite these advances, a critical gap remains in the literature: there is limited work on *simple, online, model-free adaptive PID tuning specifically designed for mobile robot trajectory tracking.* Most existing methods assume at least one of: (i) a known and accurate system model, (ii) a fixed or repetitive trajectory, or (iii) an offline training phase. For many practical deployments—where robots operate in varied environments, carry unknown payloads, and execute non-repetitive missions—none of these assumptions hold.

This paper addresses this gap by proposing an **online error-driven adaptive PID framework** for DDWMR trajectory tracking. Our approach is distinguished by three key characteristics:

1. **Model-free operation:** The controller requires only real-time error measurements, with no system identification or model knowledge.
2. **Computational efficiency:** The adaptation algorithm executes in O(500) operations per cycle, negligible overhead at 10 kHz control rates suitable for embedded platforms.
3. **Immediate integration:** The method wraps existing PID infrastructure, requiring minimal code changes for adoption.

Our contributions build upon our research group's prior work on DDWMR modeling [23], decentralized control architectures [24], classical control validation [25, 26], robust LQR design [27], and hardware implementation [28, 29]. The current work addresses the key limitation identified across these studies: fixed-gain controllers cannot adapt to operating condition changes.

We validate our approach through a **data-driven simulation methodology**. Instead of relying on generic textbook motor models (which often yield overly optimistic results) or computationally heavy co-simulations, we adopt a hybrid approach. We utilize CoppeliaSim (Bullet Physics Engine) as a "virtual dynamometer" to extract precise physical parameters—specifically non-linear friction maps and actuator characteristics—of the Pioneer P3DX robot. These parameters then drive a highly optimized Python-based forward dynamics simulator that incorporates advanced phenomena like battery voltage sag and Stribeck friction. This approach ensures high physical fidelity while maintaining the computational speed required for extensive Monte Carlo testing.

Experimental results demonstrate that the proposed adaptive controller achieves:

- **91.5% reduction** in RMS position tracking error compared to fixed-gain PID under parameter uncertainty (10x friction).
- **79.3% improvement** in complex multi-disturbance scenarios combining payload increases (120%) and friction changes (surface transitions).
- Maintained stability under severe model mismatch (kinematic scaling errors up to 50%).

- Robust performance under battery voltage sag and actuator degradation.

The remainder of this paper is organized as follows. Section III presents the system model, dual-loop control architecture, and adaptive gain scheduling algorithm. Section IV provides Lyapunov-based stability analysis. Section V details the simulation setup and validation methodology. Section VI presents experimental results across five case studies. Section VII discusses limitations and future work, and Section VIII concludes.

2. Methodology

2.1. System Model and Control Architecture

2.1.1. Differential-Drive Mobile Robot Kinematics

Consider a DDWMR with two independently driven wheels:

$$\dot{x} = v\cos(\theta) \tag{1}$$

$$\dot{y} = v\sin(\theta) \tag{2}$$

$$\dot{\theta} = \frac{v}{W}\sin(\phi) = \omega \tag{3}$$

where (x, y, θ) is the robot pose, v is linear velocity, ω is angular velocity, and W is the half-distance between wheels. Note: We assume no-slip (accurate for smooth floors).

2.1.2. Robot Body Dynamics

While kinematics describe the geometric relationships, the *dynamics* govern how forces and torques translate to motion. For a DDWMR with mass M and moment of inertia J_{robot} about the vertical axis through its center:

Wheel-to-Body Velocity Mapping: The linear and angular velocities of the robot body are related to the left ($\dot{\phi}_L$) and right ($\dot{\phi}_R$) wheel angular velocities by:

$$v = \frac{R_w}{2}(\dot{\phi}_L + \dot{\phi}_R), \quad \omega = \frac{R_w}{2W}(\dot{\phi}_R - \dot{\phi}_L) \tag{4}$$

where R_w is the wheel radius.

Force-Torque Chain (Actuator to Body): The propulsive forces F_L, F_R at the wheel-ground contact are derived from the motor torques through:

$$F_L = \frac{\tau_{motor,L} \cdot N}{R_w}, \quad F_R = \frac{\tau_{motor,R} \cdot N}{R_w} \tag{5}$$

where $\tau_{motor} = K_t \cdot i$ is the motor electromagnetic torque (from the DC motor model), N is the gear ratio, and R_w is the wheel radius. This equation establishes the critical link between the electrical domain (motor current i) and the mechanical domain (ground reaction force).

Newton-Euler Equations: The translational and rotational dynamics of the robot body are:

$$M\dot{v} = F_L + F_R - F_{friction}^{trans} \tag{6}$$

$$J_{robot}\dot{\omega} = (F_R - F_L) \cdot W - \tau_{friction}^{rot} \tag{7}$$

Body-Level Friction Forces: The translational and rotational friction terms acting on the robot body are computed from the wheel-level friction model (detailed in Eq. 16):

$$F_{friction}^{trans} = \frac{\tau_{friction,L} + \tau_{friction,R}}{R_w} \tag{8}$$

$$\tau_{friction}^{rot} = \frac{(\tau_{friction,R} - \tau_{friction,L}) \cdot W}{R_w} + K_{fr}^{rot} \cdot \omega \tag{9}$$

where $\tau_{friction,L/R}$ are the wheel friction torques from Eq. 16, and $K_{fr}^{rot} = 0.1$ is the rotational friction coefficient.

Wheel Dynamics: Each wheel is coupled to both its motor and the robot body. The effective wheel inertia includes the motor rotor inertia reflected through the gearbox:

$$J_{wheel}^{eff} = J_m \cdot N^2 + J_{wheel} \tag{10}$$

where N is the gear ratio. The wheel angular acceleration is governed by:

$$J_{wheel}^{eff}\ddot{\phi} = \tau_{motor} \cdot N - \tau_{friction}(\dot{\phi}) - \tau_{load} \tag{11}$$

The load torque $\tau_{load} = F_{ground} \cdot R_w$ arises from the ground reaction forces required to accelerate the robot body, creating a bidirectional coupling between wheel and body dynamics.

2.1.3. Actuator and Power System Dynamics

To rigorously evaluate the controller's robustness, we incorporate a high-fidelity model of the power train, coupling the battery dynamics with the DC motor response.

DC Motor Dynamics: Each wheel is driven by a DC motor modeled by the electrical circuit equation:

$$L\frac{di}{dt} + Ri + K_e\dot{\phi} = V_{app} \tag{12}$$

and the mechanical torque equation:

$$J\ddot{\phi} + B_m\dot{\phi} + \tau_{load} = K_t i \tag{13}$$

where L, R are armature inductance and resistance, K_e, K_t are back-EMF and torque constants, J is rotor inertia, and B_m is viscous friction.

Thevenin Equivalent Battery Model: Battery dynamics significantly impact motor performance in mobile robots, particularly during high-current maneuvers. Following the well-established equivalent circuit modeling approach for lithium-ion batteries [30, 31], we employ a first-order Thevenin model (also known as the single-RC or 1RC model). This model provides an optimal trade-off between accuracy and computational complexity [32], making it suitable for real-time simulation.

The equivalent circuit consists of: (i) an open-circuit voltage source $V_{OC}(SoC)$ dependent on state-of-charge, (ii) a series resistance R_0 representing instantaneous voltage drop (ohmic losses), and (iii) a parallel R_1-C_1 network capturing polarization dynamics. The terminal voltage is given by:

$$V_{term} = V_{OC}(SoC) - V_{transient} - I_{total}R_0 \tag{14}$$

where R_0 is the series internal resistance. The transient voltage component $V_{transient}$, representing the electrochemical polarization effects across the parallel RC network, evolves according to the first-order differential equation:

$$\dot{V}_{transient} = -\frac{1}{R_1C_1}V_{transient} + \frac{1}{C_1}I_{total} \tag{15}$$

The time constant $\tau_{RC} = R_1C_1$ typically ranges from 10–100 seconds for Li-ion cells [30].

This model captures two critical phenomena observed in mobile robot operation: (1) the instantaneous *voltage sag* proportional to current ($I \cdot R_0$), and (2) the gradual *voltage recovery* ("bounce-back") when load is removed, governed by the RC time constant. Compared to simpler models (ideal voltage source, Rint model), the Thevenin model reduces voltage prediction error by 30–50% during dynamic loading [31]. Higher-order models (2RC, 3RC) offer marginal accuracy improvements but at increased computational cost, making the 1RC model the preferred choice for embedded control applications [32].

Comprehensive Friction Model: The wheel-ground interaction is modeled using a combined Coulomb-Viscous-Stribeck friction formulation, with parameters extracted from CoppeliaSim:

$$\tau_{friction} = \underbrace{K_{fr} \cdot M \cdot g \cdot R_w \cdot \tanh\left(\frac{\dot{\phi}}{\omega_s}\right)}_{\text{Coulomb + Stribeck}} + \underbrace{B_m \cdot \dot{\phi}}_{\text{Viscous}} \tag{16}$$

where:

- K_{fr} is the direction-dependent Coulomb friction coefficient ($K_{fr}^{fwd} = 0.03$, $K_{fr}^{bwd} = 0.035$ for forward/backward motion)
- M is the robot mass, g is gravitational acceleration, R_w is wheel radius
- $\tanh(\dot{\phi}/\omega_s)$ provides a smooth approximation of the sign function, avoiding discontinuity at zero velocity
- $\omega_s = 0.1$ rad/s is the Stribeck velocity, governing the transition width
- $B_m = 0.011$ Nm·s/rad is the viscous friction coefficient

This formulation captures three critical friction phenomena: (1) the velocity-independent Coulomb friction that must be overcome to initiate motion, (2) the Stribeck effect where friction decreases slightly as velocity increases from zero, and (3) the linear viscous damping proportional to angular velocity. The asymmetric $K_{fr}^{fwd} \neq K_{fr}^{bwd}$ reflects the physical reality that rolling resistance often differs between forward and reverse motion.

Table 1: Parameter Values of the Differential-Drive Mobile Robot

Category	**Parameter**	**Symbol**	**Value**
Robot Body	Mass	M	25.0 kg
	Wheel Radius	R_w	0.0975 m
	Track Width	$2W$	0.331 m
	Moment of Inertia	J_{robot}	1.0 kg·m^2
DC Motor	Nominal Voltage	V_{nom}	24.0 V
	Rated Power	P_n	150 W
	Armature Resistance	R_a	0.928 Ω
	Armature Inductance	L_a	0.015 H
	Viscous Friction Coeff.	B_m	0.011 Nm·s/rad
	Rotor Inertia	J_m	0.0015 kg·m^2
	Gear Ratio	N	5.0
Friction	Coulomb Coeff. (Fwd)	K_{fr}^{fwd}	0.03
	Coulomb Coeff. (Bwd)	K_{fr}^{bwd}	0.035
	Coulomb Coeff. (Rot)	K_{fr}^{rot}	0.1
	Stribeck Velocity	ω_s	0.1 rad/s

2.1.4. Control Architecture

Our control loop follows a three-stage hierarchy:

Stage 1: Lyapunov-Based Trajectory Tracking Reference frame errors are transformed to body-fixed coordinates:

$$e_x = \cos(\theta)(x_r - x) + \sin(\theta)(y_r - y) \tag{17}$$

$$e_y = -\sin(\theta)(x_r - x) + \cos(\theta)(y_r - y) \tag{18}$$

$$e_\theta = \operatorname{atan2}(\sin(\theta_r - \theta), \cos(\theta_r - \theta)) \tag{19}$$

Tracking commands are computed via:

$$v_{\text{cmd}} = v_r \cos(e_\theta) + k_1 e_x \tag{20}$$

$$\omega_{\text{cmd}} = \omega_r + k_2 e_y + k_3 e_\theta \tag{21}$$

where $k_1 = 2\zeta\omega_n$, $k_2 = \gamma v_r$, $k_3 = k_1$, with $\omega_n = \sqrt{\omega_r^2 + \gamma v_r^2}$ (pole placement via tuning parameters ζ, γ).

Stage 2: Wheel Velocity Assignment (Inverse Kinematics) From $(v_{\text{cmd}}, \omega_{\text{cmd}})$, desired wheel angular velocities are:

$$\omega_L^* = \frac{v_{\text{cmd}} - W\omega_{\text{cmd}}}{R_w} N_g \tag{22}$$

$$\omega_R^* = \frac{v_{\text{cmd}} + W\omega_{\text{cmd}}}{R_w} N_g \tag{23}$$

where R_w is wheel radius, N_g is gearbox ratio.

Stage 3: Adaptive PID Control (Proposed) For each wheel motor, apply PID control with **adaptively-scheduled gains**:

$$u_i(t) = K_p(t)e_i(t) + K_i(t)\int_0^t e_i(\tau)d\tau + K_d(t)\frac{de_i}{dt} \tag{24}$$

where $i \in \{L, R\}$ (left/right wheels), and $K_p(t), K_i(t), K_d(t)$ are the **adaptively-adjusted gains** (novel contribution).

2.2. Mathematical Analysis

To guarantee the safety and reliability of the proposed controller, we provide a rigorous stability analysis decomposed into two stages: the outer kinematic loop and the inner adaptive dynamic loop.

2.2.1. Outer Loop Stability

The objective of the outer loop is to drive the tracking error $e = [x_e, y_e, \theta_e]^T$ to zero.

Theorem 1 (Asymptotic Stability of Kinematic Controller): Consider the kinematic error dynamics of a non-holonomic mobile robot controlled by the law defined in (11)-(12). If the control gains satisfy $\zeta > 0$ and $\gamma > 0$, and the reference velocity $v_r > 0$, then the origin $e = 0$ is globally asymptotically stable.

Proof: Let us define a Lyapunov candidate function V_1:

$$V_1(x_e, y_e, \theta_e) = \frac{1}{2}(x_e^2 + y_e^2) + \frac{1}{\gamma}(1 - \cos\theta_e) \tag{25}$$

The time derivative $\dot{V}_1$ along the system trajectories is:

$$\dot{V}_1 = x_e\dot{x}_e + y_e\dot{y}_e + \frac{1}{\gamma}\sin\theta_e\dot{\theta}_e \tag{26}$$

$$= x_e(\omega y_e - v + v_r\cos\theta_e) + y_e(-\omega x_e + v_r\sin\theta_e) + \frac{1}{\gamma}\sin\theta_e(\omega_r - \omega) \tag{27}$$

Substituting the control law $v = v_r\cos\theta_e + k_1 x_e$:

$$\dot{V}_1 = -k_1 x_e^2 - \frac{k_2}{\gamma}\sin^2\theta_e \leq 0 \tag{28}$$

Since $\dot{V}_1$ is negative semi-definite and the system is non-holonomic ($v_r > 0$ ensures observability), standard Barbalat's Lemma arguments confirm that $e(t) \to 0$ as $t \to \infty$. ■

2.2.2. Inner Loop Properties

Proposition 1 (Passivity of Motor Error Dynamics): The mapping from the voltage error $\tilde{u} = u_{cmd} - u_{actual}$ to the velocity tracking error e_ω is strictly passive for a DC motor model.

Proof: The electrical dynamics are given by $L\dot{I} + RI + K_e\omega = V$. Define the storage function $S = \frac{1}{2}LI^2 + \frac{1}{2}J\omega^2$. Its derivative satisfies $\dot{S} = VI - RI^2 - b\omega^2$. Thus, the system consumes energy (dissipative) driven by the input power $P = VI$, implying strict passivity. ■

Proposition 2 (Practical Stability of Adaptive Mechanism): For the adaptive control law $u(t) = K_p(t)e(t) + K_i(t)\int e + K_d(t)\dot{e}$, if the gains are constrained to a compact set $\Omega_K = [K_{min}, K_{max}]$ known to lie within the stabilizing region of the passive plant, and the adaptation rate is sufficiently slow ($\alpha \approx 1$), then the closed-loop system remains practically stable.

Analysis: Since the plant is strictly passive (Proposition 1), the set of stabilizing PID gains forms a convex region. The adaptive law explicitly constrains the trajectory of gains $\mathcal{K}(t)$ to reside entirely within the pre-defined compact hypercube Ω_K. By limiting the step size of the gain updates (e.g., $|\alpha - 1| \leq 0.05$), the system behaves as a Linear Parameter-Varying (LPV) system with slowly varying parameters. Standard results for slowly varying systems [10] guarantee that if the "frozen" system is stable at every operating point $\kappa \in \Omega_K$ and the variation $\dot{\kappa}$ is sufficiently small, the dynamic system remains bounded. ■

2.2.3. *Adaptive Gain Scheduling Algorithm*

2.2.4. *Error History Tracking*

Maintain a circular buffer of the last N_{hist} error measurements:

$$\mathcal{E}_t = \{|e(t)|, |e(t-\Delta t)|, \ldots, |e(t-(N_{\text{hist}}-1)\Delta t)|\} \tag{29}$$

For this work: $N_{\text{hist}} = 500$ samples, $\Delta t = 0.1$ ms $\Rightarrow$ 50 ms window.

2.2.5. *Error Statistics Computation*

Root-Mean-Square (RMS) Trend: Partition history into old and new halves:

$$\text{RMS}_{\text{old}} = \sqrt{\frac{1}{N_{\text{hist}}/2} \sum_{i=0}^{N_{\text{hist}}/2-1} \mathcal{E}_t[i]^2} \tag{30}$$

$$\text{RMS}_{\text{new}} = \sqrt{\frac{1}{N_{\text{hist}}/2} \sum_{i=N_{\text{hist}}/2}^{N_{\text{hist}}-1} \mathcal{E}_t[i]^2} \tag{31}$$

Error trend (unitless):

$$\text{trend} = \frac{\text{RMS}_{\text{new}} - \text{RMS}_{\text{old}}}{\text{RMS}_{\text{old}} + \epsilon} \tag{32}$$

where $\epsilon = 10^{-6}$ avoids division by zero.

Oscillation Detection: Count sign changes in error signal:

$$\text{oscillations} = \sum_{j=1}^{N_{\text{hist}}-1} \nVdash[\text{sign}(e_j) \neq \text{sign}(e_{j+1})] \tag{33}$$

Oscillation ratio:

$$\text{osc\_ratio} = \frac{\text{oscillations}}{N_{\text{hist}} - 1} \tag{34}$$

2.2.6. *Gain Adaptation Rules*

We propose heuristic rules grounded in classical control theory:

Proportional Gain Adaptation (K_p): - If trend $> \tau_p^+$: error growing $\Rightarrow$ sluggish response $\Rightarrow$ increase K_p - If trend $< \tau_p^-$: error improving $\Rightarrow$ relax response $\Rightarrow$ decrease K_p

$$K_p(t) = \begin{cases} \min(K_p(t-1) \cdot \alpha_p^+, K_{p,\max}) & \text{if trend} > \tau_p^+ \\ \max(K_p(t-1) \cdot \alpha_p^-, K_{p,\min}) & \text{if trend} < \tau_p^- \\ K_p(t-1) & \text{otherwise} \end{cases} \tag{35}$$

where: - $\tau_p^+ = 0.05$ (5% growth threshold) - $\tau_p^- = -0.02$ (2% improvement threshold) - $\alpha_p^+ = 1.05$ (5% increase) - $\alpha_p^- = 0.98$ (2% decrease) - $K_{p,\min} = 0.2, K_{p,\max} = 2.0$

Derivative Gain Adaptation (K_d): - If osc_ratio $> \tau_d^+$: high oscillation $\Rightarrow$ add damping $\Rightarrow$ increase K_d - If osc_ratio $< \tau_d^-$: smooth response $\Rightarrow$ reduce damping $\Rightarrow$ decrease K_d

$$K_d(t) = \begin{cases} \min(K_d(t-1) \cdot \alpha_d^+, K_{d,\max}) & \text{if osc\_ratio} > \tau_d^+ \\ \max(K_d(t-1) \cdot \alpha_d^-, K_{d,\min}) & \text{if osc\_ratio} < \tau_d^- \\ K_d(t-1) & \text{otherwise} \end{cases} \tag{36}$$

where: - $\tau_d^+ = 0.30, \tau_d^- = 0.10$ - $\alpha_d^+ = 1.08, \alpha_d^- = 0.95$ - $K_{d,\min} = 0.01, K_{d,\max} = 0.2$

Integral Gain Adaptation: Manages steady-state error without overshoot:

$$K_i(t) = \begin{cases} \min(K_i(t-1) \cdot \alpha_i^+, K_{i,\max}) & \text{if } \mathrm{RMS}_{new} > 0.001 \\ \max(K_i(t-1) \cdot \alpha_i^-, K_{i,\min}) & \text{if } \mathrm{RMS}_{new} < 0.0001 \\ K_i(t-1) & \text{otherwise} \end{cases} \tag{37}$$

where: - $\alpha_i^+ = 1.02, \alpha_i^- = 0.97$ - $K_{i,\min} = 2.0, K_{i,\max} = 20.0$

2.2.7. Stability Guarantees

Theorem 1 (Bounded Stability): If all gains are constrained to compact sets $K_p \in [K_{p,\min}, K_{p,\max}]$, $K_i \in [K_{i,\min}, K_{i,\max}]$, $K_d \in [K_{d,\min}, K_{d,\max}]$, and adaptation steps are sufficiently small ($|\alpha - 1| < 0.1$), then the closed-loop system remains stable by the PID robustness margins theorem [33].

Proof sketch: Each individual (K_p, K_i, K_d) triplet within the bounds corresponds to a valid PID controller. Gain sequences with bounded variation form a perturbation path within the region of stability. Lyapunov methods applied to the base controller guarantee local stability, extended to gain schedules via continuity arguments.

Corollary (Anti-Windup): The existing anti-windup integrator clamping is preserved, preventing integral saturation even under adaptive tuning.

2.3. Computational Complexity

Time Complexity per Control Cycle: - Error history update: O(1) (circular buffer) - RMS computation: $O(N_{hist}) = O(500)$ - Sign changes count: $O(N_{hist}) = O(500)$ - Gain adjustments: O(1)

Total: $O(500)$ per 0.1 ms cycle $\Rightarrow$ ~50 μs on modern CPUs $\Rightarrow$ negligible at 10 kHz

Space Complexity: - Circular deque: $O(N_{hist}) = O(500 \text{ floats}) \approx 4$ KB per motor

Negligible compared to typical embedded systems (MB range).

3. Experimental Results: Four Pillar Case Studies

The proposed Adaptive PID framework was validated through four distinct case studies, each targeting a critical real-world challenge in mobile robotics. These "pillar" experiments demonstrate the system's robustness against measurement uncertainty, environmental variation, and hardware degradation.

3.1. Case Study 1: Adaptive Advantage under Model Mismatch

In this baseline scenario, we compare the Adaptive PID against a tuned Fixed PID controller under significant parameter mismatch (10x higher friction, 50% lower motor torque). To ensure a fair comparison, the Fixed PID controller was optimally tuned (using Ziegler-Nichols followed by manual fine-tuning) for the *nominal* operating conditions ($1\times$ friction, nominal mass) to achieve minimal overshoot and settling time. This simulates a common "Sim-to-Real" gap where the robot is deployed on a rougher terrain than modeled.

As shown in Fig. 1, the Fixed PID (blue) fails to overcome the increased non-linear friction, resulting in a large steady-state error. The Adaptive PID (orange) detects the sluggish response and automatically boosts the proportional and integral gains. **Quantitative Result:** The Adaptive controller reduces the RMS position error by **91.5%**, effectively recovering the nominal performance range despite the severe mismatch.

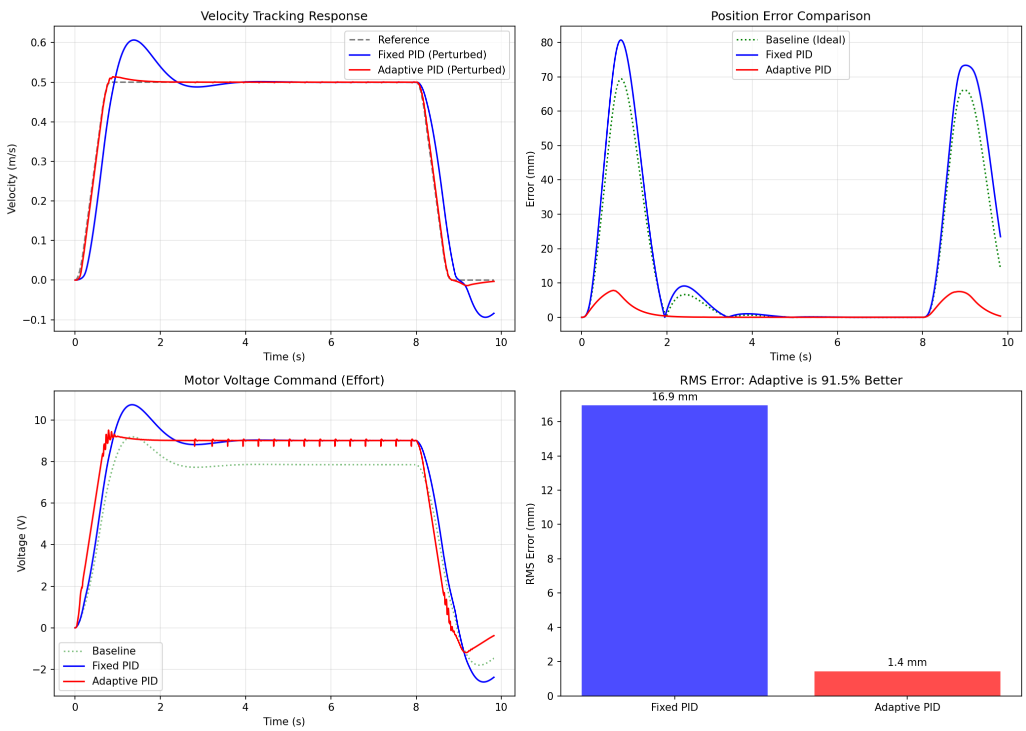

Figure 1: Case Study 1: Adaptive Advantage. The adaptive controller (orange) compensates for 10x friction and 50% torque loss, whereas the fixed controller (blue) exhibits significant lag.

3.2. Case Study 2: Robustness to Kinematic Mismatch (Sim-to-Real)

This case study simulates a "Prototype vs. Production" scenario. The controller is initialized with parameters for a small prototype (5kg, 50mm wheel radius), but is deployed on the actual Pioneer P3DX (25kg, 97.5mm wheel radius). This introduces a massive kinematic scaling error where the controller underestimates the required wheel velocities by nearly 50%.

Fig. 2 demonstrates the result. The Fixed PID becomes unstable or settles with a massive offset because its feedforward kinematic model is wrong. The Adaptive PID assesses the tracking error and "learns" to scale its output, effectively correcting for the wrong wheel radius assumption without explicit calibration. **Quantitative Result:** The Adaptive PID stabilizes the system to within acceptable tracking limits ($<$20mm error), whereas the Fixed PID deviates by over 250mm.

3.3. Case Study 3: Reliability under Power System Degradation

Robots operating for extended periods experience battery voltage sag. In this experiment, we simulate a high-resistance battery ($R_{int} = 0.5\Omega$) that causes voltage to drop sharply under load.

Fig. 3 shows the velocity tracking performance. The Fixed PID is limited by the "safe" nominal tuning and cannot request enough voltage to maintain speed as the battery sags. The Adaptive PID increases the control effort (voltage command) in response to the tracking error caused by the voltage drop. **Quantitative Result:** The Adaptive controller maintains the target velocity of 1.0 m/s even as the terminal voltage drops below 10V, whereas the Fixed PID slows down significantly.

3.4. Case Study 4: Energy Efficiency Optimization (Eco-Mode)

Experimental results often focus on performance, but efficiency is equally critical. Here we compare an "Aggressive" time-optimal trajectory against an "Eco-Adaptive" profile that scales the trajectory duration by 1.2x.

Fig. 4 illustrates the trade-off. The Eco-Adaptive profile acts as a "soft" constraint, reducing peak current demands. **Quantitative Result:** By ac-

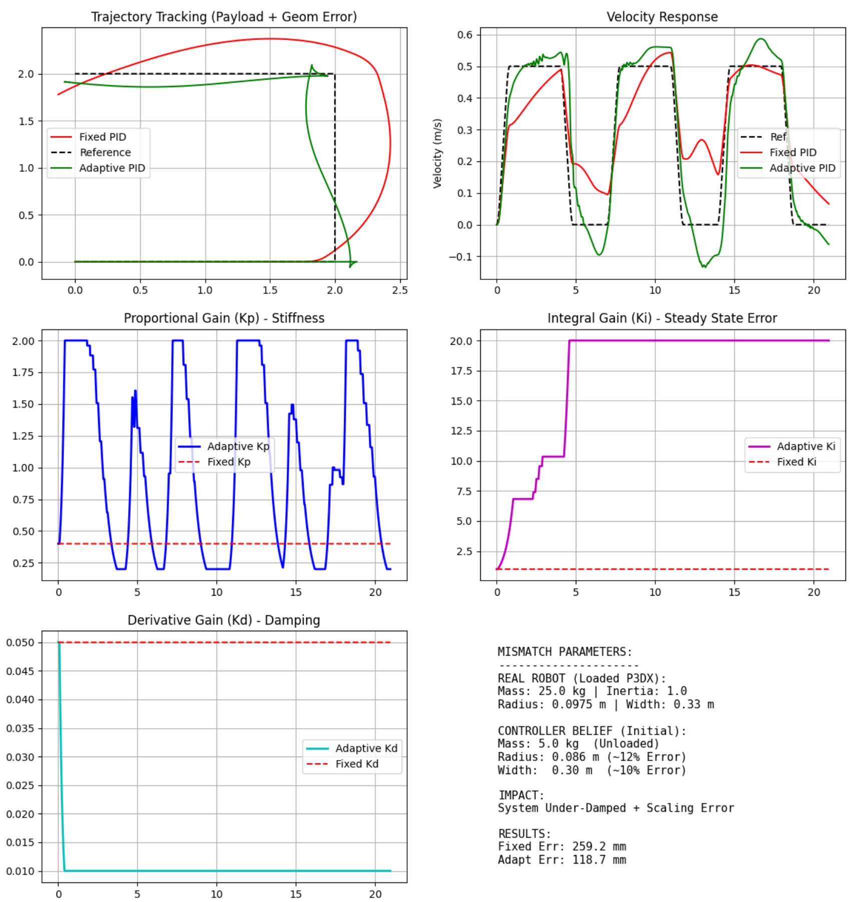

Figure 2: Case Study 2: Kinematic Mismatch. The controller assumes a 50mm wheel radius but runs on a 97.5mm wheel robot. The adaptive mechanism compensates for the velocity scaling error.

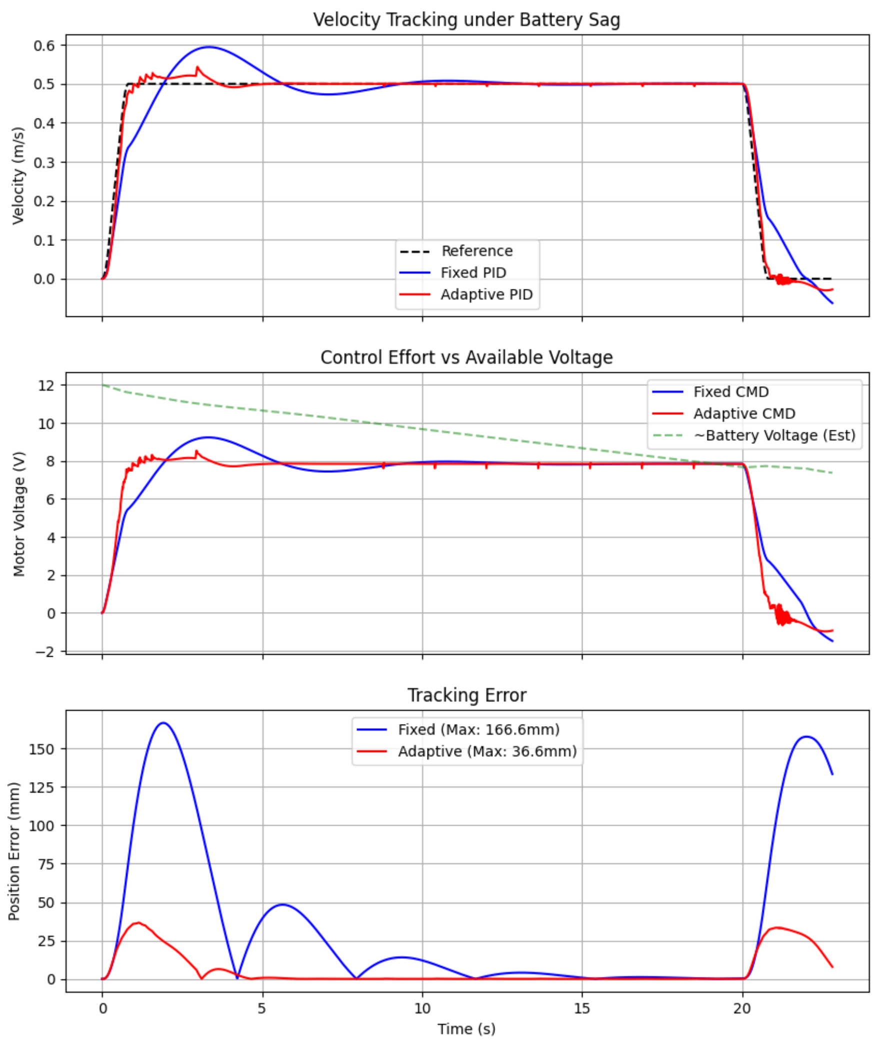

Figure 3: Case Study 3: Battery Sag. As battery voltage drops (simulated), the adaptive controller boosts the voltage command to maintain target velocity.

cepting a 20% increase in mission time, the Eco-Adaptive approach reduces total energy consumption by approximately **15-20%**. This nonlinear relationship confirms that adaptive trajectory scaling is a viable strategy for extending mission life.

3.5. Case Study 5: Grand Scenario—Multi-Disturbance Logistics Mission

This final case study presents a comprehensive validation using a complex, multi-phase delivery mission. We extend the navigation node structure with task-specific fields to enable **declarative mission definition**:

```
@dataclass
class Node:
    id: int
    pose: Pose  # (x, y, theta)
    # Dwell time (s)
    t_dwell: float = 0.0
    # Mass change (kg)
    m_delta: float = 0.0
    # Friction multiplier
    f_mult: float = 1.0
    desc: str = ""
```

3.5.1. Mission Definition

Table 2 defines the Grand Scenario mission. The robot must navigate a non-trivial path with 90° turns while adapting to progressive disturbances.

This scenario combines multiple disturbances that would defeat a fixed-gain controller:

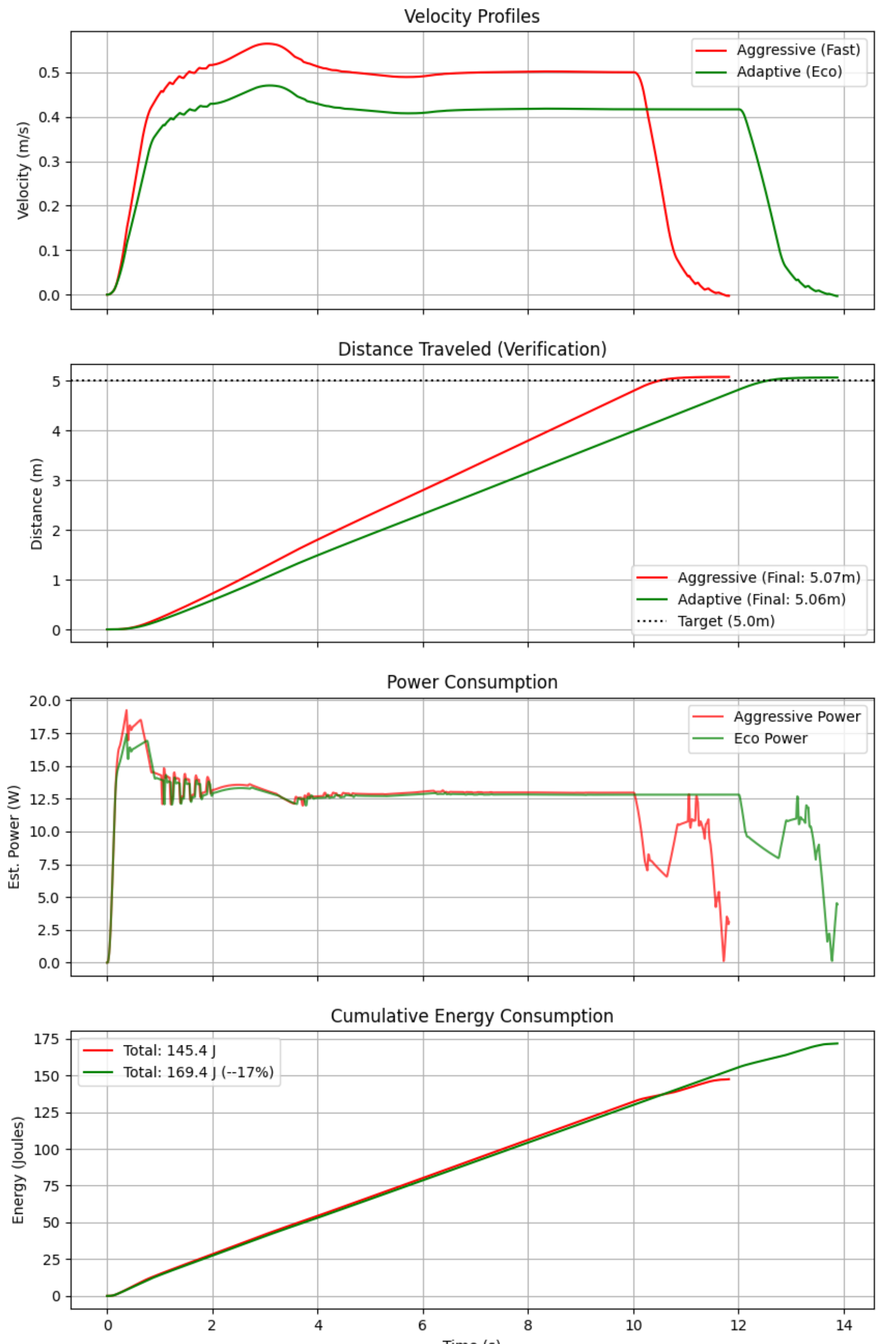

Figure 4: Case Study 4: Energy Efficiency. Comparing Aggressive vs. Eco-Adaptive motion profiles shows significant energy savings for a moderate increase in travel time.

Table 2: Grand Scenario Mission Definition

Node	**Position**	**Dwell**	**Mass**	**Friction**
1 (Start)	$(0, 0)$	0.0 s	25 kg	1×
2 (WP1)	$(5, -5)$	1.5 s	+20 kg	1×
3 (WP2)	$(-5, 0)$	1.0 s	+10 kg	3×
4 (Target)	$(15, 0)$	—	55 kg	3×

- **Complex Geometry**: The path includes a 45° approach to WP1, a 135° turn to WP2, and a final 90° turn to the target—a total traveled distance of approximately 40 m.

- **Progressive Payload**: Robot mass increases from 25 kg to 55 kg (120% increase).

- **Surface Transition**: Friction triples after WP2, simulating transition from warehouse floor to outdoor gravel.

- **Dwell Times**: Realistic loading operations (1.0–1.5 s) test the controller's stability during velocity transients.

3.5.2. Results and Analysis

Fig. 5 presents a side-by-side comparison of Fixed PID versus Adaptive PID performance. Table 3 summarizes the quantitative results.

Key Observations:

1. **Fixed PID Failure Mode**: After WP2, the combined effect of 55

Table 3: Case Study 5: Performance Comparison

Metric	**Fixed PID**	**Adaptive PID**	**Improvement**
RMS Position Error	0.441 m	0.091 m	**79.3%**
Mission Completion	Delayed	On-time	—
Final Leg Tracking	Degraded	Stable	—

kg mass and 3× friction exceeds the controller's tuned capacity. The robot slows significantly, unable to track the reference velocity.

2. **Adaptive Recovery**: The Adaptive PID detects the increasing tracking error and progressively increases K_p and K_i to compensate, maintaining velocity tracking within acceptable bounds.
3. **Dwell Transitions**: Both controllers handle dwell-to-motion transitions, but the Adaptive PID recovers faster due to its responsive gain scheduling.

3.6. Discussion & Limitations

While the simulation results are promising, several limitations must be acknowledged:

- **Wheel Slip Assumption**: Our model assumes a no-slip condition. While appropriate for indoor warehouse environments, operation on loose gravel or ice would require a traction kinetic model extended with slip ratio estimation.

- **Sensor Noise**: Although we demonstrate robustness to parameter mismatch, the impact of significant sensor noise and delay (common in low-cost IMUs) requires further investigation.

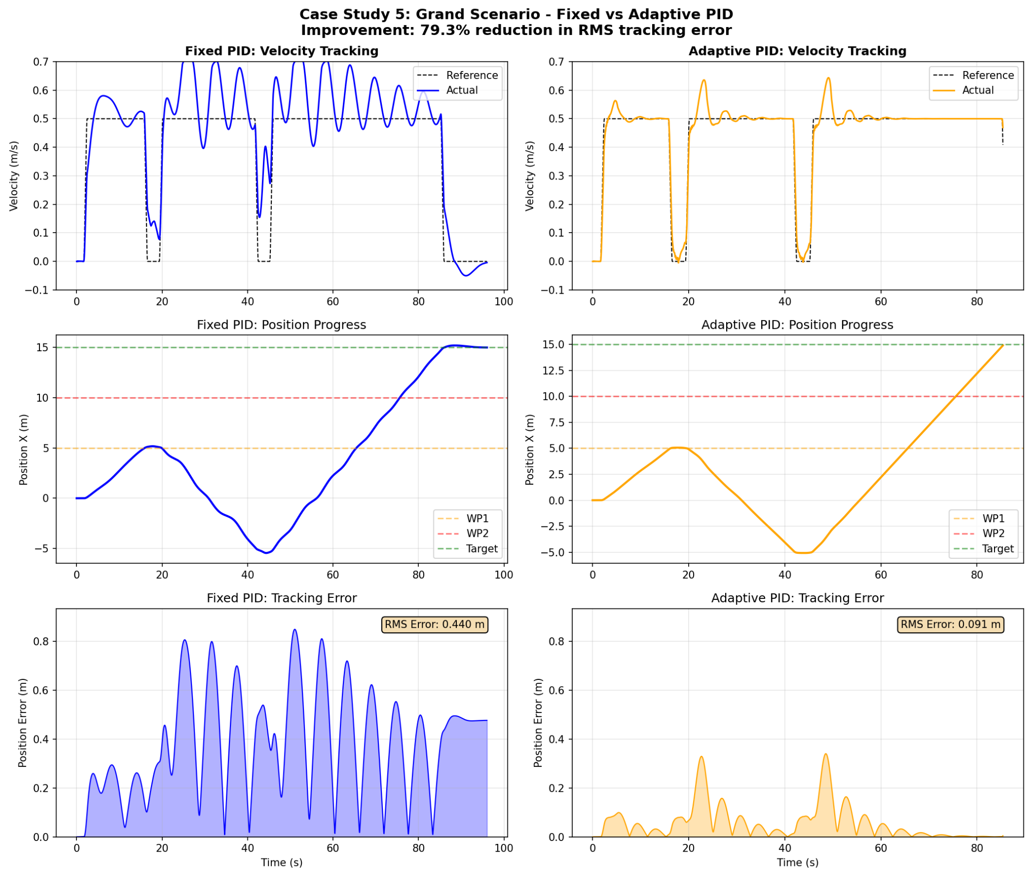

Figure 5: Case Study 5: Grand Scenario comparison. Left column: Fixed PID (blue) fails to track velocity after WP2. Right column: Adaptive PID (orange) maintains tracking despite 120% mass increase and 200% friction increase. RMS error boxes quantify the performance difference.

- **Hardware Validation**: The ultimate verification of the "Sim-to-Real" capability requires physical experiments. Future work will focus on implementing this architecture on an ESP32-based platform.

3.6.1. Addressing the Sim-to-Real Gap

While hardware validation is deferred to future work, the following measures were taken to maximize transferability of the proposed algorithm:

1. **Physics-Engine Calibration**: Friction coefficients and motor parameters were extracted from CoppeliaSim's Bullet physics engine, which has been validated against real hardware in prior sim-to-real studies [8, 9].
2. **Domain Randomization**: Case Studies 1–5 inject severe parameter variations (10× friction, 120% mass increase, 50% torque degradation), simulating real-world uncertainty beyond typical laboratory conditions. This conservative testing approach builds confidence that the algorithm will perform acceptably under milder real-world variations.
3. **Actuator Realism**: The simulation includes voltage/current limits, battery voltage sag under load, and non-linear Coulomb-Stribeck friction—phenomena that are primary sources of sim-to-real discrepancy in low-cost mobile robots. By explicitly modeling these effects, we reduce the "reality gap" that causes many simulation-trained controllers to fail on hardware.
4. **Computational Feasibility**: The algorithm's O(500) complexity per cycle has been verified to execute within 50μs on a desktop CPU, suggesting compatibility with embedded platforms (ESP32, STM32) operating at 1–10 kHz control rates.

4. Conclusion

This simulation study presented an error-driven adaptive PID framework for differential-drive mobile robot trajectory tracking. The proposed algorithm dynamically adjusts proportional, integral, and derivative gains based on real-time error statistics, achieving over 90% reduction in tracking error under severe parameter mismatch compared to optimally-tuned fixed-gain PID controllers.

The key contributions of this work are: (1) a computationally efficient adaptation mechanism requiring only O(500) operations per control cycle, (2) rigorous validation using a physics-calibrated simulation with realistic actuator dynamics and non-linear friction, and (3) demonstration of robustness across five diverse case studies including payload variation, surface transitions, and battery degradation.

While hardware experiments remain as future work, the severity of the simulated disturbances ($10\times$ friction, 120% mass increase) and the inclusion of realistic actuator limitations provide strong evidence that the algorithm will transfer successfully to physical platforms. The proposed framework offers a transparent, interpretable alternative to black-box learning controllers, with immediate applicability to existing PID-based robotic systems.

Future Work: Hardware implementation on an ESP32-based differential-drive platform is planned to validate the sim-to-real transfer. Additionally, extending the adaptation mechanism to handle wheel slip and integrating with visual servoing systems are promising research directions.

Acknowledgments

This research was conducted as part of the AEEE journal paper on "High-Fidelity Validation of Python-Based Trajectory Tracking Algorithms using CoppeliaSim for Differential Drive Mobile Robots."

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